Date: 20/02/2026

Practice question on the GHCR
Question 1.

```
You are required to build a Docker image from a Dockerfile and publish
it to
GitHub Container Registry (GHCR), and then pull it on another system.

Task Requirements:
1. Create a simple Dockerfile for a Python Flask application.
2. Build the Docker image locally and tag it appropriately for GitHub
Container Registry (ghcr.io).
3. Authenticate Docker with GitHub Container Registry using a
Personal Access Token (PAT).

4. Push the Docker image to ghcr.io/<your-github-
username>/<image-name>:v1.

5. Verify that the image is successfully uploaded in your GitHub
Packages section.
6. On a different system (or after removing the local image):
• Login again to GHCR.
• Pull the same image.
• Run the container and verify the application is working.
```

Prerequisite:
Go to the

Step 1: Make a dockerfile according to the question.

```
FROM python:3.11-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install -r  requirements.txt
COPY . .
EXPOSE 5000
CMD ["python","app.py"]
```

dockerfile

```
FROM python:3.11-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install -r  requirements.txt
COPY . .
EXPOSE 5000
CMD ["python","app.py"]
```

Step 2: Build dockerfile using:
docker build -t ghcr.io/<github_username>/<any_image_name> .

```
docker build -t ghcr.io/sakura-winter/practiceflask
```

```
sumitkumarmehta@sumits-MacBook-Air PythonProject % docker build -t ghcr.io/sakura-winter/practiceflask .
[+] Building 2.2s (11/11) FINISHED                                                                  docker:desktop-linux
```

```
sumitkumarmehta@sumits-MacBook-Air PythonProject % docker build -t ghcr.io/sakura-winter/practiceflask .
[+] Building 2.2s (11/11) FINISHED                                                                  docker:desktop-linux
 => [internal] load build definition from dockerfile                                                                0.0s
 => => transferring dockerfile: 180B                                                                                0.0s
 => [internal] load metadata for docker.io/library/python:3.11-slim                                                 2.1s
 => [auth] library/python:pull token for registry-1.docker.io                                                       0.0s
 => [internal] load .dockerignore                                                                                   0.0s
 => => transferring context: 2B                                                                                     0.0s
 => [1/5] FROM docker.io/library/python:3.11-slim@sha256:0b23cfb7425d065008b778022a17b1551c82f8b4866ee5a7a200084b7e2eafbf   0.0s
 => => resolve docker.io/library/python:3.11-slim@sha256:0b23cfb7425d065008b778022a17b1551c82f8b4866ee5a7a200084b7e2eafbf   0.0s
 => [internal] load build context                                                                                   0.0s
 => => transferring context: 93B                                                                                    0.0s
 => CACHED [2/5] WORKDIR /app                                                                                       0.0s
 => CACHED [3/5] COPY requirements.txt .                                                                            0.0s
 => CACHED [4/5] RUN pip install -r  requirements.txt                                                               0.0s
 => CACHED [5/5] COPY . .                                                                                           0.0s
 => exporting to image                                                                                              0.0s
 => => exporting layers                                                                                             0.0s
 => => exporting manifest sha256:cdf9d9783f8a1a473e40540c6b587675b692e8d84e8396085e7ebc650e2b8019                   0.0s
 => => exporting config sha256:236d1028bd488f51eb920df41607da6d6e5215ec3c00b5ea54b85237bdac8c0a                     0.0s
 => => exporting attestation manifest sha256:ab243bc730d060e9b8153d5c2d321b195495e965492032c50f93d4a14bfe5f2c       0.0s
 => => exporting manifest list sha256:823150b1586c9ff2dc46000eece172854c228162555a057e85869539ac3f27d4              0.0s
 => => naming to ghcr.io/sakura-winter/practiceflask:latest                                                         0.0s
 => => unpacking to ghcr.io/sakura-winter/practiceflask:latest                                                      0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/0qdb9lvzsc3evxf52dzll98bn
```

Step 3: Now Docker Login via ghcr.io using
docker login ghcr.io
username: <github_username>
Password: <PAT>

```
sumitkumarmehta@sumits-MacBook-Air PythonProject % docker login ghcr.io
Authenticating with existing credentials... [Username: sakura-winter]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
```

Step 4: Push to the GHCR

docker push ghcr.io/<github_username>/<image_name>

```
docker push ghcr.io/sakura-winter/practiceflask
```

```
sumitkumarmehta@sumits-MacBook-Air PythonProject % docker push ghcr.io/sakura-winter/flaskpython
Using default tag: latest
The push refers to repository [ghcr.io/sakura-winter/flaskpython]
tag does not exist: ghcr.io/sakura-winter/flaskpython:latest
sumitkumarmehta@sumits-MacBook-Air PythonProject % docker push ghcr.io/sakura-winter/practiceflask
Using default tag: latest
The push refers to repository [ghcr.io/sakura-winter/practiceflask]
09fc16548375: Mounted from sakura-winter/lol
e9e50e44f49d: Mounted from sakura-winter/lol
7174b34ca655: Mounted from sakura-winter/lol
bdace1c2060b: Mounted from sakura-winter/lol
1b2f2b74a58c: Mounted from sakura-winter/lol
5d3dcd8f65e1: Pushed
3ea009573b47: Mounted from sakura-winter/lol
93304c4769c2: Mounted from sakura-winter/lol
05c5a2414322: Mounted from sakura-winter/lol
latest: digest: sha256:823150b1586c9ff2dc46000eece172854c228162555a057e85869539ac3f27d4 size: 856
```

Step 5: Confirm on github repo and pull from github

Before pulling delete the local image created using:

```
docker rmi <imagename>
```

```
sumitkumarmehta@sumits-MacBook-Air PythonProject % docker rmi ghcr.io/sakura-winter/practiceflask
Untagged: ghcr.io/sakura-winter/practiceflask:latest
Deleted: sha256:823150b1586c9ff2dc46000eece172854c228162555a057e85869539ac3f27d4
```

Now Pull Docker image using:
docker pull <image_name>

```
sumitkumarmehta@sumits-MacBook-Air PythonProject % docker pull ghcr.io/sakura-winter/practiceflask
Using default tag: latest
latest: Pulling from sakura-winter/practiceflask
Digest: sha256:823150b1586c9ff2dc46000eece172854c228162555a057e85869539ac3f27d4
Status: Downloaded newer image for ghcr.io/sakura-winter/practiceflask:latest
ghcr.io/sakura-winter/practiceflask:latest
```

Now confirm using:

```
docker images
```

```
sumitkumarmehta@sumits-MacBook-Air PythonProject % docker images
```

i Info → U In Use

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
bash:latest	e320b40b14ab	26.7MB	7.36MB	
docker/welcome-to-docker:latest	c4d56c24da4f	22.9MB	6.35MB	U
flaskpython:latest	9c64b18a7354	238MB	52.2MB	U
ghcr.io/sakura-winter/lol:latest	82451c27740c	238MB	52.2MB	U
ghcr.io/sakura-winter/practiceflask:latest	823150b1586c	238MB	52.2MB	U
httpd:latest	dd178595edd6	207MB	47.8MB	U
mysql1:latest	ce8332ed5d12	1.06GB	228MB	U
nginx:latest	13310a9cc1de	258MB	64.1MB	U
sakurawinter/ubuntusample:v1.1	067708b231d9	451MB	116MB	
ubuntu:latest	cd1dba651b30	141MB	30.8MB	U
ulb:latest	195916186bb2	451MB	116MB	U
welcome-to-docker:latest	250e3657631c	441MB	122MB	U